

		[1] ans
7(a)	$R = \frac{\rho L}{A}$ $= \frac{1.68 \times 10^{-8} \times 15.0}{\pi(0.07^2 - 0.04^2) / 4}$ $= 2.52 \times 10^{-7} \Omega \text{ m}^2 \div 2.59 \times 10^{-3} \text{ m}^2$ $= 9.73 \times 10^{-5} \Omega$	[1] exp for area [1] sub [1] ans
(b)(i)	$R = \left\{ \left[\left(\frac{1}{30} + \frac{1}{12} \right)^{-1} + 4.5 \right]^{-1} + \frac{1}{18} \right\}^{-1}$ $= 7.6 \Omega$	[1] working
(ii)	Current in 18 Ω, = Emf $\div$ 18 Ω = 6.0 $\div$ 18 = 0.33 A	[1] ans

JURONG JUNIOR COLLEGE
PHYSICS DEPARTMENT
2018 JC2 Preliminary Examinations
9749 H2 Physics Paper 3
Suggested Solutions with Markers' Comments

Qn	Suggested solution	Remarks
(iii)	Total current = $\text{Emf} \div \text{Total resistance, } R$ $= 6 \text{ V} \div 7.6 \, \Omega$ $= 0.79 \text{ A}$ Current through $4.5 \, \Omega = 0.79 \text{ A} - 0.33 \text{ A}$ $= 0.46 \text{ A}$ Power dissipated in $4.5 \, \Omega = I^2 R \text{ \{or } V^2/R\}$ $= 0.46^2 \times 4.5$ $= 0.95 \text{ W}$	[1], total current [1], current through $4.5 \, \Omega$ [1], ans
2(a)(i)	1. Any gas is made up of very large number of molecules	[1]